



DARREN K. WOODLAND, JR.

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drexel-research-garden

EDUCATION

Drexel University, Philadelphia, PA

Ph.D. in Digital Media (Jan. 2022 – Present)

Research Focus: *Research-creation methodology, Black epistemologies, sonic interaction design, critical technical practice, physical-digital entanglement*

Dissertation: "Sonic Speculocultural Technopoiesis: Black Speculative Tradition as Modification for Embodied Practice," Drexel University. Emil Polyak, Chair

Graduate Minor in Computer Science (Sep. 2022 – 2024)

Focus: Software design, user interfaces, machine learning, audio programming

North Carolina State University, Raleigh, NC

M.A.D. in Experimental Media Arts (Aug. 2018 – Dec. 2020)

Research Focus: Sonic interaction design, deep listening, extended realities

Thesis: "On Soundscape: Listening as Interaction," North Carolina State University, 2020. Dr. Mark Searce, Chair

Graduate Certificate in Digital Humanities (May 2019 – Dec. 2020)

Focus: Games in social contexts, ludic networks, game design, game studies

University of South Carolina, Columbia, SC

B.A. (Dual) in Media Arts & Art Studio (Aug. 2013 – Dec. 2017)

Media Arts: Animation, narrative writing, game design

Art Studio: Graphic design, illustration, drawing

PUBLICATIONS

Peer-Reviewed Publications

Emil Polyak, Kathi Martin, Rghad Balkhyoor, Arefeh Ahmadi, **Woodland, Darren**, and Varun Mahadev (2024). "Plumes and Pixels: Bridging Digital and Traditional Art in SciArt Education". In: *SIGGRAPH Asia 2024 Educator's Forum*. SA '24. Tokyo, Japan: Association for Computing Machinery. ISBN: 9798400711367. DOI: [10.1145/3680533.3697072](https://doi.org/10.1145/3680533.3697072). URL: <https://doi.org/10.1145/3680533.3697072> **[Co-Author]**

Darren K. Woodland (2024). "Motifs of the Future Metamorphosis: Afrofuturism in Media Arts and Design". In: *Proceedings of the Media Arts and Design Conference*. United States: Media Arts and Design Association **[Author]**

Non-Peer Reviewed Publications

Darren K. Woodland (2022). *Cybernetic Thrills: Black Ice VR*. Audiokinetic Blog. Blog post discussing the development and design of the Black Ice VR experience. URL: <https://blog.audiokinetic.com/> **[Author]**

RESEARCH | CREATIVE COLLABORATIONS | WORKS IN PROGRESS

Robotics, Automation, and Dance (RAD) Lab, Philadelphia, PA

Researcher/Artist in Residence, Dr. Amy LaViers (2025 – Present)

Collaborating on empirical research at the intersection of embodied movement, emotion perception, and human-robot interaction. Contributing practice-based perspectives and digital media methods to inquiry into how emotional states are notated and denoted and become legible in movement. Supported by NSF-funded research on machine perception of human body language.

A River Wears What We Carry, Philadelphia, PA

Sound & Interaction Designer, Drexel University, Academy of Natural Sciences, TTF Watershed Partnership, PAR Recycle Works (2025 – Present)

Site-specific performance exploring entanglements between human, digital, and water-based systems through wearable e-waste sculptures, sonic sensors, and live dance. Responsible for soundscape design and sensor-based interaction systems that create a real-time feedback

loop between the ecological flow of Philadelphia waterways and the bodies of performers. Collaborative team includes Steve Vásquez Dolph (Drexel), Ryan Strand Greenberg (Academy of Natural Sciences), Talia Greene (lead artist), and Micah Lockman-Fine (choreographic score).

BioreactAR, Drexel University × Purdue University

AR/MR Developer & Development Supervisor (2025 – May 2026)

Converted an inherited large-scale 50L bioreactor model, originally developed for VR training and simulation at NC State University, into an augmented/mixed reality classroom multi-user training application for the Meta Quest 3. Led a small development team on AR passthrough interaction design and implementation. Collaborative project between Drexel University (Emil Polyak) and Purdue University (Dr. Xinyu Zhang and Dr. Jeremy Roberts).

CONFERENCES | TALKS | PRESENTATIONS

Presenter (Accepted) "Material Synthesis Composition: Speculocultural Technopoiesis as a Framework for Human-Material-AI Co-Creativity," Co-Creativity in Music, Sound, and AI: Improvisation, Interaction, Composition, University of California, Riverside – Culver Center of the Arts, 2026

Presenter (Accepted) Society for the Neuroscience of Creativity Conference (*SfNC2026: Liberty to Create*), Penn MindCORE & FringeArts, Philadelphia, PA, 2026

Presenter "Research: In Searching," Westphal Creates – Drexel University Westphal College of Media Arts and Design Faculty and Student Research Showcase, 2025

Presenter "Afrofuturists Motifs of Metamorphosis and Transfiguration in Media Arts and Design," Atlantica Conference, Seen from Lagos: African Re-Groundings Symposium, 2024

Presenter "Motifs of the Future Metamorphosis: Afrofuturism in Media Arts and Design," Media Arts and Design Conference, 2024

Presenter "Echoes of Harmonic Threads," Ars Electronica, Expanded Animation Symposium, 2023

GRANTS | AWARDS | FELLOWSHIPS

Design Award AIGA of SC InShow 22 Awards for *Pop! Art Book Covers*, 2017

PERFORMANCES | EXHIBITION | CREATIVE WORKS

Digital Portfolio: darrenwoodlandjr.com

Art Exhibition *Digital Twin: Transforming Human Experience Through Digital Rendering*, 2025

Contributing artist. Works presented: *Flight of the Navigator*, *Canopy*, and *The Aviator*.

Interactive Installation *BioreactAR*, RAD Lab, Drexel University, 2025–May 2026

Augmented reality classroom education simulation integrating a physically-rendered bioreactor model with user-experience interaction and user-interface to make function and processes understandable and learnable

Interactive Installation *Mnemosi for WOMB: The Black Wealth Project*, Rankin Scholar, 2024

Real-time interactive projection sculpture in which audience voice recordings triggered generative audiovisual growth of a central figure and accumulating starfield

Performance *Home is in Me*, Colorado Conservatory of Dance, 2024

Designed and developed animations, audio systems, and real-time Unreal Engine visuals for live stage production

Performance *Grililef*, Mandell Theater, Drexel University, 2024

Developed motion-derived animation systems serving as real-time visual backdrop for three dance works exploring distinct stages of grief

VR Cinema *Black Ice VR*, SIGGRAPH Immersive Pavilion, 2023

Showcased at the SIGGRAPH Immersive Pavilion; cyberpunk VR experience released on Pico VR platform

VR Cinema *Black Ice VR*, Bucheon International Fantastic Film Festival, 2023

Showcased at BIFAN; cyberpunk virtual reality experience

VR Cinema *Black Ice VR*, SXSW XR Experience, 2022

Showcased at the SXSW immersive showcase; cyberpunk virtual reality experience

VR Application *Barnstormers: Determined to Win*, MxReality Lab, Raleigh, NC, 2020

Created high-fidelity character models from historical references using Reallusion Character

Creator and iClone for early development version

Graphic Design *Pop! Art Book Covers*, AIGA of SC InShow 22 Awards, 2017

Award-winning series recognized at the AIGA South Carolina InShow

Animated Short *Scientific SC*, Knowitall.org, 2017

Educational animated short on the life of a prominent South Carolinian scientist

TEACHING EXPERIENCE

Drexel University, Philadelphia, PA

Doctoral Fellow (January 2022 – Present)

Instructor of Record

DIGM 553/753: Experimental Digital Media Lab (Graduate)

Spring 2026 Research-Creation, Epistemic Tools, Physical-Digital Entanglement

GMAP 395: Advanced Game Design and Production

Summer 2025 Particle Systems, Proceduralism, Modularity, Technical Art

DIGM 131: Introduction to Scripting for the Digital Content Creation Pipeline

Fall 2024 Python, Maya, Unity, Tool and Pipeline Development

GMAP 395: Advanced Game Design and Production

Fall 2024 Particle Systems, Proceduralism, Modularity, Technical Art

GMAP 395: Advanced Game Design and Production

Summer 2024 Particle Systems, Proceduralism, Modularity, Technical Art

GMAP 345/CS 345/DIGM 545: Game Development Foundations (Graduate/Undergraduate

Cross-List)

Fall 2023 Frameworks, Unity, Scripting, Game Design Process, Play

GMAP 345/CS 345: Game Development Foundations (Undergraduate Cross-List)

Fall 2023 Frameworks, Unity, Scripting, Game Design Process, Play

GMAP 395: Advanced Game Design and Production

Summer 2023 Particle Systems, Proceduralism, Modularity, Technical Art

GMAP 231: Scripting for Game Design (2 sections)

Summer 2023 Unity, C#, Game Design Frameworks, Scripting Design Patterns

GMAP 260: Overview of Computer Gaming

Spring 2023 Games, History of Games, Video Game History, Fundamentals of Computer Games

GMAP 345/CS 345/DIGM 545: Game Development Foundations (Graduate/Undergraduate Cross-List)

Spring 2023 Frameworks, Unity, Scripting, Game Design Process, Play

GMAP 345/CS 345/DIGM 545: Game Development Foundations (Graduate/Undergraduate Cross-List)

Winter 2023 Frameworks, Unity, Scripting, Game Design Process, Play

GMAP 395: Advanced Game Design and Production

Winter 2023 Particle Systems, Proceduralism, Modularity, Technical Art

GMAP 231: Scripting for Game Design (2 sections)

Summer 2022 Unity, C#, Game Design Frameworks, Scripting Design Patterns

Teaching Assistant

DIGM T580/GMAP T380: Games for Health and Simulation (Graduate/Undergraduate Cross-List)

Spring 2023, Instructor: Dr. Paul Diefenbach

Grading assignments, equipment setup for sensors and body tracking, in-class assistance with game design and course knowledge, office hours

GMAP 395: Advanced Game Design and Production

Spring 2022, Instructor: Kenneth Chen

Grading, in-class student assistance, assignment examples, office hours

ANIM 145: Realtime Visualization (Undergraduate)

Spring 2022, Instructor: Daniel Rose

In-classroom assistance, grading, student support

GMAP 345: Game Development Foundations (Undergraduate)

Winter 2022, Instructor: Tom Burdak

Grading, remote student assistance, full assignment examples

North Carolina State University, Raleigh, NC

Graduate Teaching Assistant (August 2018 – May 2019)

ADN 319: Animation I (Undergraduate)

Spring 2019, Instructor: Patrick Fitzgerald

Assisted in teaching industry-standard video editing and animation software and workflows along with animation principles

CURRICULUM DESIGN

Drexel University, Philadelphia, PA

Complete Curriculum Development: DIGM 553/753: Experimental Digital Media Lab

Instructor of Record (Spring 2026)

Developed comprehensive curriculum from inception as instructor of record for a newly introduced course in the redesigned Drexel Digital Media M.S. program, working from a bare-bones syllabus with no existing course materials. Designed an 11-week seminar/studio hybrid structured around research-creation methodology, integrating critical theory, physical-digital entanglement, and experimental media production. Course is open to all graduate students across the Westphal College of Media Arts & Design, with particular emphasis on first- and second-year M.S. students developing independent practices and M.S. thesis.

Course Redesign and Development: GMAP 345/DIGM 545/CS 345: Game Development Foundations

Instructor of Record (2022-2025)

Assisted in the initial redesign of the undergraduate GMAP 345 course, then served as lead developer to further redesign the curriculum for improved alignment within the game development sequence (GMAP 231, 345, 395). Delivered the redesigned curriculum as instructor of record across undergraduate and graduate cross-listed sections (DIGM 545, CS 345). Course focuses on frameworks, Unity development, scripting methodologies, game design processes, and play theory.

Complete Curriculum Development: GMAP 395: Advanced Game Art

Instructor of Record (2022-2025)

Developed comprehensive curriculum from inception as instructor of record following departure of previous instructor with no existing course materials. Designed advanced technical

game art workflows including particle systems, procedural generation, modular design principles, and technical art production pipelines. Course strategically positioned as capstone in the game art sequence, building upon foundational skills from GMAP 231 and 345.

Collaborative Curriculum Development: DIGM 131: Introduction to Scripting for the Digital Content Creation Pipeline

Instructor of Record (Fall 2024)

Co-designed new foundational course with two fellow PhD students to replace the original introductory scripting curriculum. Led development of syllabus design and course materials, serving as primary instructor for the inaugural course offering. Course introduces Python scripting, Maya integration, Unity workflows, and tool development within digital content creation pipelines.

EMPLOYMENT | CONTRACTS | INTERNSHIPS

Drexel University, Philadelphia, PA

Perforce Administrator Assistant (January 2024 – Present)

Manage Perforce version control server, facilitating workflow between students, faculty, and IT infrastructure

Develop documentation and learning resources to streamline version control adoption

Develop, implement, and maintain support structures for students and faculty using Perforce

University of North Carolina School of the Arts, Media and Emerging Technology Lab, Winston-Salem, NC

Art Director and Technical Artist (April 2021 – October 2021)

Led art direction and asset development for a VR project, creating or compiling 200+ assets

Integrated spatial audio and sound design into the production pipeline

Colorado Conservatory of Dance, Denver, CO

Freelance Multimedia Artist, Contract (January 2024 – July 2024)

Designed and developed animations and audio elements for a live stage dance production

Created real-time visuals in Unreal Engine and crafted motion-based animations

Integrated and edited multimedia content using DaVinci Resolve to enhance production quality

North Carolina State University, IntelliMedia Group (Center for Educational Informatics), Raleigh, NC

Digital Artist and Graduate Research Assistant (May 2019 – December 2020)

Led asset pipeline development, designing, modeling, and animating 2D/3D assets

Created assets for educational games and VR training applications

University of South Carolina, Department of Information Technology, Columbia, SC

Digital Artist, Contract (May 2018 – July 2018)

Designed and illustrated a large-format learning map for university-wide IT training

South Carolina Educational Television, Columbia, SC

Digital Media Intern (January 2017 – May 2017)

Produced an animated short on the life and achievements of a prominent South Carolinian scientist

PROFESSIONAL DEVELOPMENT | MEMBERSHIPS | SERVICES

Professional Memberships

The Society for Neuroscience of Creativity (SfNC)

Member (April 2026 – Present)

Research Certifications

CITI: Collaborative Institutional Training Initiative

Conflicts of Interest (COI), Social Behavior Valid 2024 – 2028

Responsible Conduct of Research (RCR), Social Behavior Valid 2024 – 2027

Human Subjects Research (HSR), Social Behavior Valid 2025 – 2028